

RealTrack



ADVANCE OBJECT TRACKING TECHNICAL SPECIFICATION



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All-in-One, Stand-Alone Analytics Software

- Rapidly add analytics to any system, VMS, PSIM or NVR software.
- Operates as a stand-alone server or as a plug-in application for several VMS and AI Cameras.
- Integrate with any IP camera or VMS via RTSP.
- State-of-the-art algorithms for tracking people, vehicle types, bags, or use-case-defined objects.
- Full analytics metadata stream provided for forensic search or other value-added applications.
- Deployable on Intel x86 and ARM hardware.
- Customisable out-of-the-box HTML5 UI (review via Chrome on Windows, Linux, Mac, Android).

RealTrack is an all-in-one analytics server and reference design that enables the rapid integration of AI tracking and classification technologies with third-party systems, such as VMSs, NVRs, and PSIM systems.

In standalone mode, RealTrack can stream video from a range of input sources and generate events in a range of formats.

Designed from the ground up to be easy to use and integrate.

With video streaming, event alerting and an HTML5 configuration interface out-of-the-box. Simple integrations can be completed in less than a week.

ANALYTIC & AI FEATURES

							
Zones	Tamper Detection	Object Tracker	Presence	Counting	Counting Lines	Hand Object Tracker	Logical Rules
							
Meta Data	3D Calibration	Appear	Disappear	Direction	Dwell	Aggressive Behaviour	Fall Detection
							
Enter	Exit	Stopping Filter	Speed Filter	Abandoned Object	DL Skeleton Tracker	DL People Tracker	DL Object Tracker

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BEST-IN-CLASS ANALYTICS

Over 1,500,000 channels of RealTrack analytics have already been deployed on PC and embedded platforms. RealTrack can upgrade any compatible IP camera, VMS, NVR, or PSIM with a best-in-class analytics solution with minimum integration effort.

WEB-BASED CONFIGURATION INTERFACE

A HTML5 web-based interface supports the configuration of detection rules and notifications on desktop and mobile devices, with the option for OEM customization and interface skinning.

EASY INTEGRATION

Easy integration with a wide range of VMS, NVR and PSIM platforms over HTTP, TCP or email. RealTrack's simple token-based message format provides relevant metadata to your platform. Enabling real-time analytics alerts, metadata overlay and forensic search capabilities from within the native VMS environment.

SYSTEM

Supported OSs	Ubuntu 22.04 LTS, Windows 11
Supported Browsers	Chrome (version 130 and above)
Supported Clients	PC, Android, Mac
Supported Platforms	Intel / AMD x86, ARMv8
Supported Video Formats	H.264, H.265, MPEG-4

BASIC EVENT RULES

Event Sources	Analytics Rules, Loss of Signal, Interval (heartbeat), Schedule, HTTP
Event Actions	E-mail, TCP, HTTP, Arm/Disarm

HARDWARE RECOMMENDATIONS



Designed to run on many different combinations, allowing the flexibility to deploy in any scenario. Refer to our website for more details

REALTRACK LICENCES

RealTrack:	Tripwires, zones and tamper. Complete rule set including logical rules and a wide range of deep learning trackers
RT-Behavior:	Aggressive Behavior, Fall, Hands Up and Hand Object Interaction
Custom:	Custom Tracking Solutions - Call us.

ADVANCED AI ANALYTICS ENGINE

State-of-the-art deep learning classification and people tracking engines for reduced false alarms and accurate business intelligence metadata.

LOGICAL RULES

Simple to use logical rules engine, allowing the coupling of rules to create advanced behaviour detection. Create bespoke alarms and events.

OPEN STANDARD API

Fully documented, open standard, REST API and metadata streams, for simple integration into a range of customized monitoring scenarios and applications. Facilitates simple event reporting to real-time metadata streaming for forensic search.

STAND-ALONE ANALYTICS SERVER MODE

Deploy as a stand-alone analytics server, streaming video from cameras or VMSs directly, or as a reference design for deeper integrations.

SUPPORTED ANALYTICS

Deep Learning Object and People Tracking	Deep learning trackers for highly accurate detection and classification of people, hands, vehicle types and objects. Optimised for business intelligence and metadata generation.
Event Rules	Intrusion detection, Tamper, Enter & Exit filters, Appear & Disappear filters, Stopped filter, Dwell filter, Direction & Speed filters, Counting, Abandoned & Removed object filter, Zones & lines, 3D calibration, Tailgating filter, Aggressive Behaviour, Fall Detection, Logical rules.
Object Detection, Tracking and Classification	Robust object tracking engine tracks through occlusion. Deep learning classification for reduced false positive rate. Multiple modes permit optimization for intrusion detection.
Object Counting	Accurate people and vehicle type counting, even in dense scenes.
Tamper	Camera Tamper Alert, Loss of Video Input Connection Alert.

SUPPORTED INTERFACES

Video Input Method	RTSP, Files (avi, mp4, mpg)
Output Data Format	Events (TCP, HTTP, SMTP, VMS / NVR / PSIM Client) Metadata (JSON)
Web Interface	HTML5 via embedded web server, No plugins necessary
API	RESTful API for 3rd party integration and customization, ONVIF profile S and T

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PROCESSOR ARCHITECTURES	SUPPORTED PLATFORMS	SUPPORTED VIDEO FORMATS
  	   	  

GPU HARDWARE

*Hardware shown has been tested with Ubuntu, maximum channel numbers are shown. Other solutions are available.
Input stream configuration: 640x480, 15FPS, and high bitrate for good image quality

APPROPRIATE CPU REQUIRED

Tracker	DL Object / People	DL Skeleton	Hand Object Interaction	Behaviour (Aggressive, Fall)
GPU	Number of Channels			
GTX 1660 Super	22	18	22	8
RTX A1000	32	30	33	14
RTX 3050	44	43	36	28
T1000	17	17	17	6
RTX 3070	60	45	60	30
RTX A2000 (12GB)	42	45	45	25
RTX 4060	77	81	68	40
RTX 4060 ti	101	95	108	75
RTX A4000	70	70	70	40
RTX A5000	130	130	130	60

ARM HARDWARE

*Hardware shown has been tested with Ubuntu, maximum channel numbers are shown. Other solutions are available.
Input stream configuration: 640x480, 15FPS, and high bitrate for good image quality

Tracker	DL Object / People	DL Skeleton	Hand Object Interaction	Behaviour (Aggressive, Fall)
Device	Number of Channels			
Jetson Nano	1	1	-	-
Jetson Xavier NX	5	5	6	3
Jetson Orin 4GB	5	5	5	2
Jetson Orin 8GB	8	8	8	4